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Module 7 Introduction

Lesson 7: Maximum Likelihood Inference

Reading: Maximum Likelihood Inference Reading 6/100%

Video: Maximum Likelihood Inference - Part 1 6/100%

Video: Maximum Likelihood Inference - Part 2 6/100%

Practice Assignment: Maximum Likelihood Inference Quiz - Part 1 Submitted

Practice Assignment: Maximum Likelihood Inference Quiz - Part 2 Submitted

Lesson 2: Bayesian Inference

Module 7 Summative Assessment

Module 7 Summary

Your grade: 100%

Your best: 100%

Your highest: 100%

Maximum Likelihood Inference Quiz - Part 2

Next item →

Instructions

Review Learning Objectives

1. True or False: In maximum likelihood inference, the likelihood and the probability of observing the data given the parameters are the same.
COOCH

☐ True

☒ False

Ready to review what you've learned before starting the assignment? I'm here to help.

Help me practice

Let's chat

☐ Correct

Correct! While related, likelihood and probability are different concepts. Likelihood is a function of the parameters given the data, while probability is a function of the data given the parameters.

Submitted

Attempts

Nov 16, 11:51 PM '24

Unsaved

2. True or False: Maximum likelihood estimators are always unbiased.

☐ True

☒ False

Your grade
To pass you need at least 80%. We keep your highest score.

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1 / 1 point

3. Why is the logarithm often used in practical maximum likelihood estimation?

☐ To convert non-linear models into linear ones.

☐ To increase the computational speed of the algorithm.

☐ To ensure that the likelihood is always a positive number.

☒ To simplify calculations, especially for products of probabilities, and to prevent numerical underflow.

☐ Correct

Correct! Logarithms simplify multiplication to addition and help in handling very small numbers.

4. What is a likelihood function in maximum likelihood estimation?

☐ A linear function that represents the relationship between variables in a model.

☐ A function that calculates the maximum possible likelihood for any model.

☒ A function that represents the likelihood of the observed data given different values of model parameters.

☐ A function used to predict new data points in a model.

☐ Correct

Correct! The likelihood function expresses how likely the observed data is for various parameter values.

5. What characteristics of maximum likelihood estimators is particularly advantageous?

☐ They can be calculated without any data.

☐ They require no assumptions about the underlying distribution of the data.

☒ They are consistent, meaning they converge to the true parameter values as sample size increases.

☐ They always provide the exact true parameter values, regardless of sample size.

☐ Correct

Correct! Consistency is a key property of MLE, ensuring reliability with larger samples.

6. What does it mean if the maximum likelihood estimate of a parameter lies on the boundary of the parameter space?

☐ It implies that the likelihood function is uniformly distributed.

☐ It guarantees that the model is correctly specified.

☒ It suggests that the true value might be at the extreme of the parameter range, indicating possible model misspecification.

☐ It means that no maximum likelihood estimate exists.

☐ Correct

Correct! Boundary estimates can indicate extreme values or issues with the model.

7. In maximum likelihood estimation, what is an asymptotic property?

☐ A rule that determines the minimum sample size required for MLE.

☐ A property that ensures estimators are unbiased at any sample size.

☒ A property that describes the behavior of an estimator as the sample size approaches infinity.

☐ A characteristic that guarantees the convergence of an estimator in a finite number of steps.

☐ Correct

Correct! Asymptotic properties are about how estimators behave with very large sample sizes.

8. How can overfitting occur in the context of maximum likelihood estimation?

☐ If the maximum likelihood estimation is performed without a likelihood function.

☒ If the model is too complex, it might fit the random noise in the data rather than the underlying pattern.

☐ When the model is too simple to capture the complexities of the data.

☐ When the sample size is too large for the model.

☐ Correct

Correct! Overfitting happens when complex models capture noise instead of the actual signal.

9. What is the role of the Fisher information in maximum likelihood estimation?

☒ It measures the amount of information that an observable random variable carries about an unknown parameter.

☐ It calculates the probability of the observed data.

☐ It is used to directly compute the maximum likelihood estimator.

☐ It determines the maximum value of the likelihood function.

☐ Correct

Correct! Fisher information quantifies how much the data tells us about the parameter.

10. What advantage does maximum likelihood estimation have over methods of moments?

☐ MLE does not require any assumptions about the underlying data distribution.

☒ MLE often leads to more efficient estimators with desirable statistical properties like consistency and normality.

☐ MLE can be applied without any data.

☐ MLE is computationally simpler and faster than methods of moments.

☐ Correct

Correct! MLE tends to produce efficient and statistically desirable estimators.

https://www.coursera.org/learn/llnoss-tech-statistical-learning/assignment-submission/LLp71maximum-likelihood-inference-quiz-part-2/view-feedback

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